Lung Cancer: An interplay between Behavioural & Psychosocial Factors with Early Clinical Markers

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***Abstract*—Lung Cancer is till date a terrifying disease with very low survival rates. This report carries out statistical analysis exploring interactions between behavioural and psychosocial interactions with Lung Cancer clinical markers. Data preprocessing and descriptive statistics was applied on dataset. Pearson's correlation was applied and displayed ALCOHOL_CONSUMING (r = 0.2944) and ALLERGY (r = 0.3336) had close correlation with Lung cancer. Further analysis using inferential statistics was applied on 3 models: Base Model, Interaction Model and Full Model with likelihood ratio chose Interaction Model as best model ($\chi^2 = 13.24$, $p = 0.0102$) and saw strong interactions on the behavioural and psychosocial factors with lung cancer with ALCOHOL_CONSUMING, ALCOHOL_ANXIETY and PEER_PRESSURE leading. Anova test was carried out on different age groups categorising them by risk profiles, gender and behavioural and psychosocial markers supporting Risk_Level and Smoking_Alcohol influenced lung cancer. Bayesian statistics using Watanabe–Akaike information criterion (WAIC) and Leave-one-out cross-validation (LOO-CV) with results from WAIC supporting base model with Elpd of –92.70 while LOO-CV also supported Interaction model as best model with Elpd of 182.96.**

I. Introduction

Lung cancer, a killer carcinoma is the most frequent cause of cancer deaths experienced amongst cancer patients globally; with tracking rate being usually tedious to compile due to data collation and quality implications, has seen its mortality rate stay relatively constant as a major killer having a 28% rate in the 2010s [1] and a decade later 22% in 2020s [2]. This form of cancer is often attributed to smokers [3], with traces of being dated to the massive marketing of cigarettes before World War II [8], however in recent reports, 1 in 4 lung cancer patients have never smoked before [4]. Lung cancer could also be contracted through exposure to harmful air pollutants [9], type 2 diabetes [10] and even hereditary [11]. Lung cancer severity is broken into stages from 0 to IV with IV being the most severe form of it [5]. Many efforts are being invested to detect it early enough especially at stage 0 from wedge resection with a high recovery rate [5]; in stage I, a more invasive lobotomy is required to remove the affected lung lobe; in stage IV often referred to as the Metastatic stage, treatment is usually very painful and mainly experimental especially stage IVB. What is even more worrisome is the post survival rate after a successful treatment with only 76% surviving the first 24 months post treatment [6] and an even lower 30% do not make it past 5 years [7].

The psychological struggles of which include depression, anxiety and social isolation is a key characteristic observed in lung cancer patients with over 3 in 10 patients [12] experiencing and exhibiting these struggles. The factors that typically trigger these have been linked to long stays in hospitals receiving treatment, loss of strength in carrying out physical activities and abandonment from friends during the treatment periods [13]. Analysis on typical behavior of lung cancer patients revealed that 99 in 100 Chinese nationals started smoking due to peer pressure [14]. Behavioral patterns such as smoking, alcohol consumption does not seem to slow down even after patients have been informed of their diagnosis with about 4% of patients still continuing with smoking and 18% of peers still smoking; also 43% of patients still continued alcohol consumption and over 71% of their family members doing so as well [15].

The interplay between lung cancer and psychosocial and behavioural factors was supported by the UK Biobank which carried out research concluding that without the smoking variable, psychological distress increased lung cancer risks [16]. Even post recovery, only 8 in 100 patients were living healthily and meeting properly planned food requirements.

II. Literature Review

A. Behavioural Risk Factors

In a study done comparing colorectal cancer to lung cancer, 78 in 100 lung cancer patients were obese and the higher the age of the individual, the lower the low muscle mass with a 75% likelihood [17]. The average age for lung cancer is between 40 to 80 years old with median of 63 years old [18]. The likelihood of death increases after the age of 60 for patients with low fruit diet and 80 for smokers [19].

Between the 1990s and 2021, there is an overall increase in the death rate for lung cancer with a percent change of 62.05% for smokers with women having a higher increase of 88% and men 57% [19]. Further research was carried out to investigate the relationship of smoking to bodily functions in lung cancer patients, it was observed that an abnormally high metabolism rate and stronger connectivity is present in the individuals [20]. Due to insufficient healthcare infrastructure and workers, proposals to deprioritise smokers (people with option luck) from health treatments are being proposed [21] to promote fairness. In a study comparing behavioral habits such as smoking, alcohol consumption, physical inactivity and similar lifestyle creeps for 16,000, 48% of them performed these habits have a strong influence leading to lung cancer [24].

Dietary patterns have been proven to influence lung cancer inversely with studies showing that people consuming fish, red meat, fresh fruits and vegetables had a lower risk of having lung cancer while processed meat such as bacon increased the likelihood of lung cancer [23]. Further studies on natural remedies like daily consumption of green tea, reduced odds of lung cancer by 8.2% [25]. Quite a number of reports have come out in the past evidencing the association of vitamin b12 with lung cancer. However, more recent studies seem to be inconclusive [26], suggesting no direct relationship between them when taken in an optimal range.

B. Psychological and Sociological Burden

With 25% of lung cancer patients experiencing stigma and increased depression blame themselves and regret many actions [22], the psychological and social factors are quite key areas to be studied.

Anxiety and depression, common stressors are typically exhibited by lung cancer patients and in comparison, to other terminal diseases such as covid-19 [27]. Even in post-treatment, anxiety biomarkers are still very present in recovering patients scaring them with PTSDs that affect cognition and lowers quality of life [28]. Research also further suggests that these stress related burdens are as a result from withdrawal from ever smokers [29].

Survivors who have lived the longest have been supported emotionally and socially by peers and family and this enabled them pull through [30].

C. Research Question

Exploring various papers, there are none that explore the interactions between the factors such as anxiety and peer pressure which are under psychosocial and other factors like smoking and alcohol consumption which are under behavioral habits especially across age and gender demographics. With this in mind I propose the null hypothesis to be:

H_0 (null hypothesis): *There is no significant interaction; the impact of psychosocial and behavioral habits on lung cancer is the sum of their independent events and no integration exists.*

I also propose the alternative hypothesis to be:

H_1 (alternative hypothesis): *There is a significant interaction; implying that individuals that experience psychosocial stress and behavioral habits have a high risk of lung cancer and this integration is moderated by age and gender.*

My primary research question aims to answer: whether the interplay between psychosocial factors and behavioral habits increases likelihood of cancer and if it is moderated by age and gender.

III. Research Methods

A. Dataset Description

The dataset used for this experiment was obtained from Kaggle with cc0 license for the dataset. The dataset has been vetted for ethical compliance and no markers for patients identification is in the dataset. The dataset consists of 309 records and 16 distinct features which are compiled to address research in lung cancer patients. Data pruning and purging was carried out on the dataset to remove duplicates, applying binary categorisation on categorical variables which reduced the dataset to 276 entries. The dataset had mainly categorical variables. The only continuous variable in this dataset was age.

B. Descriptive Statistics

The variable 'age' has a relatively symmetrical distribution with a mean of 62.91 years and median of 62.50 years. The age data varied from 21 to 87 years. The standard deviation (SD) of 8.38 and variance of 70.21 indicates moderate variability of the dataset. 50% of individuals fall within 11.25-year range due to the first quartile (Q1) of 57.75 and third quartile (Q3) of 69.00.

The categorical variables had a healthy mix of patients' characteristics. The gender distribution is nearly balanced with more male participants comprising 51.45% compared to females for 48.55%. A modest skew was also observed for behavioural factors: smokers – 54.35%, alcohol – 55.07%. Clinical markers such as yellow fingers, coughing, shortness of breath is observed in more than 50% of participants. Finally, the outcome 'Lung Cancer' is present having a substantial majority of 86.23%. The primary categorical variables can be found below.

TABLE I. Primary Categorical Variables

S/N	Variable	Unique Values	Value	Frequency	Percentage (%)
1	SMOKING	2	1/0	150/126	54.35/45.65
2	ANXIETY	2	1/0	137/139	49.64/50.36
3	PEER_PRESSURE	2	1/0	140/136	50.72/49.28
4	ALCOHOL_CONSUMING	2	1/0	152/124	55.07/44.93
5	LUNG_CANCER	2	1/0	238/38	86.23/13.77

Looking at the distribution by individual features, the behavioural and psychosocial variables indicate some interesting patterns for this dataset. 84.02% of smokers and 87.33% of never smokers have lung cancer while other variables have larger differences such as individuals with anxiety, yellow fingers and peer pressure have lung cancer rates > 90% while the absence is much lower.

Also, looking into the interplay between psychosocial factors: anxiety and peer pressure, respondents with Yes_Yes have a high lung cancer rate of 96.43%. Same experience is noted for behavioural factors: smoke and alcohol consumption (Yes_Yes) have as well a high rate of cancer at 96.2%. Respondents with none of them had a much lower rate at 71.7%.

C. Correlation Analysis

An initial analysis was carried out on the dataset using Pearson and Spearman's Correlation to get the top correlations among the variables. The top observed correlation was between ANXIETY and YELLOW_FINGERS variables with r = 0.5583 which indicates a strong link between these 2.

TABLE II. Top 3 Correlations

Variable A	Variable B	Correlation
ANXIETY	YELLOW_FINGERS	0.5583
SWALLOWING_DIFFICULTY	ANXIETY	0.4788
ALCOHOL_CONSUMING	GENDER	0.4343

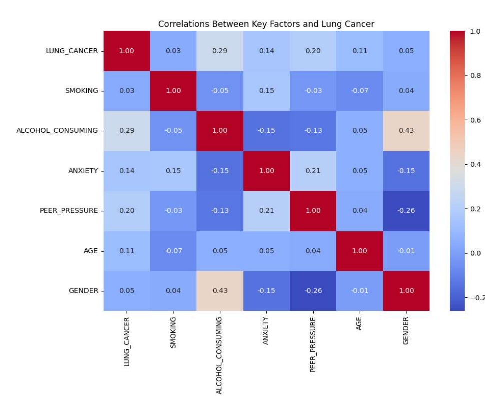

Fig. 1: Correlation between Key Factors and Lung Cancer

On direct correlations with Lung Cancer, ALCOHOL_CONSUMING ($r = 0.2944$) and ALLERGY ($r = 0.3336$) display a moderate positive correlation with lung cancer. The psychosocial factors ANXIETY (0.1443) and PEER_PRESSURE 0.1951 diverge with lower correlations which indicates that for this dataset, behaviour factors are greater associated with lung cancer.

IV. RESULTS

Three models were explored for both Inferential and Bayesian statistics. The Base Model for just the main effects, the Interaction Model that adds the Psychosocial and Behavioural Interactions and the Full Model adding age and gender.

A. *Inferetial Statistics Using Logistic Regression*

Three regression models were fitted; I used likelihood ratio tests and it indicated that the addition of psychosocial and behavioural factors (Interaction Model) improved the model to produce a much better fit ($Chi^2 = 11.05$, $df = 8$, $p = 0.1991$) compared to the base model ($Chi^2 = 13.24$, $df = 4$, $p = 0.0102$). The Interaction model has a log-likelihood = -79.17, AIC = 180.35, BIC = 220.17, $R^2 = 0.2842$ showed ALCOHOL_CONSUMING has a strong positive association ($\beta = 4.1713$; Odds Ratio = approximately 64.8; $p < 0.001$); interaction effects: ALCOHOL_ANXIETY ($\beta = –2.2142$; OR = 0.1092; $p = 0.0235$) and PEER_PRESSURE ($\beta = 1.4326$; OR = 4.1894; $p = 0.0268$) were significant. Model performance had a classification accuracy = 86%, AUC = 0.87.

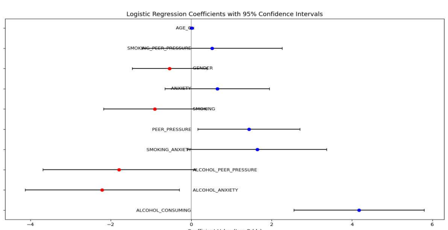

Fig. 2: Logistic Regression Coefficient Plot

1) *Inferential Statistics Using Confidence Interval (CI)*

Comparing the Base with the Interaction Model, taking ALCOHOL_CONSUMING displays a coefficient (β) = 4.1713, 95% CI = (2.5477, 5.7949), corresponding odds ratio OR = (12.78, 328.63) and 65x more likely. ALCOHOL_ANXIETY $\beta = –2.2142$, CI = (–4.1307, –0.2977), OR CI = (0.0161, 0.7425) indicating impact of alcohol consumption is moderated significantly by anxiety. The PEER_PRESSURE $\beta = 1.4326$, CI: (0.1647, 2.7004), OR CI: (1.1791, 14.8856) with 4x more likely supports the insights that psychosocial factors are interacting with behavioural factors to increase risk of lung cancer.

Other main effect variables: GENDER, SMOKING, ANXIETY, and AGE_C, the CI include 0 (or OR include 1) which implies that these variables are not significantly affecting lung cancer risk for this model.

2) *Inferential Statistics Using Anova Test*

To further look into group-level differences, I introduced 3 Risk Levels based on lung cancer rates using KMeans clustering on the dataset to standardise it with a random state of 42 which has an [$F (2, 273) = 6.2666$, $p = 0.0022$] with further validation using Tukey's HSD tests showed there was a major difference in High and Low Risk groups (mean difference = 0.1281, $p = 0.0342$), comparing by GENDER, the difference was not much [$F (1, 274) = 0.7914$, $p = 0.3745$]. Another validation on behavioural patterns with groups: 'Alcohol Only', 'Both', 'Neither', 'Smoking Only' was quite significant [$F (3, 272) = 8.9771$, $p < 0.001$]. Comparing 'Alcohol Only' and 'Neither' showed they differ significantly, 'Both' and 'Smoking Only' also differ significantly meaning lung cancer is directly influenced by these pairs.

B. *Bayesian Analysis using Bayesian Logistic Regression*

Bayesian Logistic Regression was applied on the same 3 models with 4 chains, 2000 draws and 5000 tunes. Using Widely Applicable Information Criterion (WAIC) the base model was preferred with a score = -92.57. I observed the difference between base model and the others were less than 1 which prompted me to apply Leave-One-Out Cross-Validation (LOO-CV) to validate the results due to LOO-CV being preferred because of the standard estimate error [31]. LOO-CV supported the observation from Logistic Regression and favours the Interaction model (rank = 0, elpd_loo=182.96, ploo = 7.079819) suggesting interactions improved performance. The standard error gotten from the difference between Interaction and Full = 0.8 is low suggesting no meaningful difference between both of them.

TABLE III. MODELS EVALUATION WITH LOO-CV

Model	Rank	Elpd_loo	P_loo	Dse
Interaction	0	182.96	7.07	0.00
Full	1	183.77	9.58	1.96
Base	2	185.19	6.02	1.76

TABLE IV. MODELS EVALUATION WITH WAIC

Model	Elpd_waic
Base	–92.70
Full	-91.46
Interaction	-91.66

Applying Approximate Bayes Factor had a Bayes Factor of 13,645 comparing Interaction with Bayes and an extremely high value comparing interaction with Full Model indicating the full model has no predictive gain beyond the interactions alone. Interaction Model performance had AUC = 0.8224 and classification accuracy = 87%.

1) *Bayesian Analysis with 95% Credible Intervals*

ALCOHOL_CONSUMING Mean coefficient 2.0248 (95% CI: [1.1925, 2.8921]); odds ratio 8.33 (95% CI: [3.30, 18.03]), PEER_PRESSURE (Interaction Effect) Mean coefficient 1.1733 (95% CI: [0.4092, 1.9606]); odds ratio 3.50 (95% CI: [1.51, 7.10]) showing both a strong effect and significant contribution. ANXIETY, AGE_C, SMOKING, and GENDER did not show significant effects.

2) *Bayesian Analysis using MCMC Trace Diagnostics*

Trace plots for SMOKING, ALCOHOL_CONSUMING, ANXIETY, PEER_PRESSURE, SMOKING_ANXIETY showing kernel density estimates for multiple chains and converge to a unimodal distribution showing it is stable. The progression over iterations mixes well and bounded by a common mean and reached stationary distribution.

V. CONCLUSION AND FUTURE WORKS

This paper explored Lung cancer and its interplay between Psychosocial and Behavioural Factors interaction with Lung Cancer Markers. Collectively, likelihood ratio test, Anova test, Bayes Regression upheld the alternative hypothesis that lung cancer is not just the sum of independent psychosocial and behavioural factors. The Bayesian Analysis showed that factors such as alcohol consumption and peer pressure increase the risk of lung cancer substantially with odds ratio exceeding 3 and 10 respectively. The likelihood ratio test confirmed this with $\chi^2 = 13.24$, $p = 0.0102$ identifying similar factors as well as addition of ALCOHOL_ANXIETY. The ANOVA results also supported the evidence with significant differences detected in lung cancer by Risk_Level and Smoking_Alcohol supported that those with high peer pressure or those who smoke and consume alcohol have clear lung cancer risks in comparison with others.

The analyses all reject the null hypothesis (H_0) that the impact of psychosocial and behavioral factors on lung cancer operates independently.

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